Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	All-in-one laser micromachining: high-precision structuring and polishing in one system			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	Grinding/Polishing Machine Tools	Surface Treatment	Surface Property Modification Technologies
Concept of Proposed Technology	Core concept & main function: • Integrated process chain combining ablation, cleaning and polishing processes on a single ultrashort pulse laser system for application on metal materials • Prevention of positioning errors resulting from re-clamping by all-in-one laser machining system • Enabling locally selective polishing based on surface melting with high lateral and vertical precision • Control of the surface properties (gloss level, structure fidelity, surface roughness, melting depth) by flexible adjustment of the polishing parameters • Enabling polishing of thin metal sheets by controlled heat input Potential use and application • Production and finishing of, e.g., mold inserts and final products			

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Rationale for Joint R&D Project	Ultrashort pulse (USP) laser micromachining is increasingly applied industrially for precise fabrication of functional components by micro structuring, micro drilling, grooving or surface functionalization. Typically, the surface roughness after USP ablation is in the range of Sa = 0.4-2 µm. Further post processing like cleaning and polishing is often required while maintaining a high level of fidelity of the microstructures. USP laser polishing in combination with a previous micro processing step offers a fully integrated digital USP process chain in one clamping and rework can be massively reduced. Furthermore, polishing results in terms of surface properties can be flexibly adjusted by controlled heat input during the melting process. The implementation of an integrated process chain for USP laser microfabrication requires Fraunhofer ILT's broad process expertise of all process steps in combination with the knowledge of the customer's needs, industrial manufacturing capabilities and know-how of efficient manufacturing systems of Korean SLEs. Thus, an international partnership starting with a joint R&D project is essential to combine complementary expertise and to implement advanced laser microfabrication in an industrially viable production setup.
Key Words	1. Ultrashort pulse laser ablation and polishing 2. Integrated process chain 3. Tool surface finishing 4. Mold insert processing

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-5)	End of the Project	Level (6-7)
R&D Objectives	• Process development for high-precision laser structuring and polishing using a wavelength of 1030 or 515 nm and a pulse duration of <10 ps • USP laser micro polishing process based on melting of a thin surface layer • Optical and mechanical design and machine integration of the pilot system • Demonstration of a combined ablation and polishing process on a single ultrashort pulse laser system installed at the Korean SME's site, applied to a defined industrial use-case (e.g. mold insert surface finishing)			
R&D Target Specifications	Pilot system for integrated USP ablation, cleaning and polishing with the following process performance: • Surface roughness after polishing < 50 nm • Polishing rate > 50 cm²/min • Adjustable melting depth for surface polishing from <1 µm up to 20 µm • Polishing under air atmosphere without visible surface oxidation (coloration) • Materials: metals			

R&D Plan	**Key tasks:**
	• On-going exchange of requirements (application, i.e., quality requirements of the parts; and system technology), samples, results and knowledge
	• Development of suitable polishing process parameters tailored to the application
	• Definition of the required system technology (laser beam parameters, laser beam propagation & deflection, peripheral units)
	• Machine integration and testing at SME in Korea
	Methodologies:
	• Laser process development for USP polishing includes variation of (exemplary, especially relevant parameters):
	  ○ Burst mode: number of pulses in burst
	  ○ Pulse and burst repetition rate (kHz-GHz)
	  ○ Focal shape and spot size
	  ○ Scan speed
	  ○ Track distance
	• Measurement of surface roughness, surface morphology, melting depth
	Expected milestones:
	• M1 (after month 6): Selection of suitable laser system
	• M2 (after month 27): Laser processes developed for specified application
	• M3 (after month 36): Machine integrated at SME and performance validated
	Timeline:
	• Approximated project duration: 36 months
	• Year 1:
	  ○ definition of a suitable laser beam source
	  ○ determination of suitable system technology (beam deflection and scanning system, focusing unit) on basis of basic process development
	• Year 2:
	  ○ Process improvement using the defined system technology
	  ○ Process adjustment regarding the specified application
	  ○ Setup of the basic machine at SME site
	• Year 3:			

	○ Finalization of machine at SME, final integration and adjustment, testing

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer ILT contributes extensive experience in ultrashort pulse laser polishing, holding two national patents in this field. The institute has successfully implemented selective 5-axis laser polishing on steel injection molding tools and demonstrated full polishing of 2.5D tungsten carbide tool surface for powder pressing applications using a galvo z-shifter system. Based on this foundation, Fraunhofer will support the process and system development for the Korean SME. Expected contributions and outputs: • Development of combined structuring and polishing process using Fraunhofer's existing laboratory infrastructure • Optical and mechanical design of a pilot system suitable for replication of the combined process • Provision of process parameters, guidelines, and training materials • Support in process replication and functional validation at the Korean SME Capabilities: • 420 employees with more than 11,000 m2 of usable floor space • >15 high precision USP laser machines (USP laser power up to 1 kW, wavelength from IR to DUV, pulse duration down to 50 fs, flexible beam shaping with >100 beamlets) • Several inline process monitoring sensors as OCT, WLI, confocal sensors, etc. • Well-equipped in-house metrology (SEM, AFM, LSM, WLI) Contributions: • IP on ultrashort pulse based precision laser ablation processes • IP on development and setup of optical processing heads for tailored intensity distributions or multi beam systems • Comprehensive collection of system technology for laser-based precision manufacturing • Access to ILT's supplier and industry partner network
Expected R&R of Korean SME Partner	The Korean SME will coordinate the project and be responsible for implementing and demonstrating the dual-mode laser process under near-industrial conditions. Based on ILT's system and process design, the SME will assemble the pilot system and evaluate its performance within the target use-case. Expected contributions and outputs: • Project coordination and definition of industrial use-case requirements • Provision of sample material • Procurement and integration of laser and optical components based on Fraunhofer's design • Assembly and commissioning of the pilot system at the SME's site • Process replication and functional part manufacturing in a near-industrial environment • Plan for commercialization and exploitation including international IP protection

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Pilot-level laser system at Korean SME for combined ablation and polishing of mold inserts demonstrated under near-industrial conditions • Elimination of downstream polishing steps for cost and time savings in mold tool production • Enabling of new part functionalities through selective laser polishing (e.g. hybrid gloss/matte textures, functional microfeatures), creating new market opportunities for advanced mold design and surface engineering • Joint generation of exploitable process know-how and potential IP • Invention or patents for this technology • Further public funded and bilateral projects for further industrialization of the technology towards TRL 9

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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